ADVANCED PHOTO VOLTAIC THERMO CHEMICAL HYDRATION REACTOR

Bachelor of technology

In

Department of Mechanical Engineering

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Abstract

- In the face of escalating global water scarcity, the imperative for sustainable water solutions has reached a critical juncture. Solar stills, harnessing the inexhaustible energy of the sun to produce freshwater, emerge as a beacon of hope in mitigating this urgent challenge. This project is dedicated to the meticulous optimization of solar still efficiency through a thorough exploration of design modifications and operational strategies.
- By conducting systematic experimentation and employing rigorous analysis, we delve into the intricate dynamics of various factors influencing solar still performance. These factors encompass the geometric configuration of the still and the judicious selection of absorber materials. Our endeavor is to strike an optimal balance between maximizing freshwater output and minimizing energy consumption.
- At the heart of our inquiry lies the recognition of Tiruchirappalli's unique environmental context. By tailoring our approach to suit the specific conditions of this urban landscape, we aim to develop solar stills that are not only highly efficient but also eminently practical for deployment in Tiruchirappalli and analogous settings.
- The insights garnered from this study hold profound implications for addressing water scarcity challenges both locally and globally. By pushing the frontiers of solar-driven desalination technology, our project charts a course toward the realization of sustainable water provision solutions that are not only efficacious but also adaptable to diverse environmental exigencies, including urban landscapes like Tiruchirappalli.

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INTRODUCTION

Access to clean and potable water is unequivocally recognized as a fundamental human right and an indispensable element of sustainable development. Despite this acknowledgment, numerous regions worldwide, including densely populated urban hubs such as Tiruchirappalli (Trichy), grapple persistently with the multifaceted challenges of water scarcity and the absence of reliable access to safe drinking water. Aligned with the global imperative to confront and ameliorate these pressing issues, this project is steadfastly focused on the conception, development, and optimization of a solar still prototype meticulously tailored for deployment in Tiruchirappalli.

At its core, the project seeks to harness the abundant and renewable energy of the sun to effectuate the production of freshwater through the proven principles of evaporation and condensation. By harnessing sustainable energy sources and drawing upon locally available materials, the envisioned prototype endeavors to proffer a sustainable, economically viable solution to the endemic water scarcity plaguing urban landscapes.

Through the execution of this project, our overarching aspiration is to actively contribute to the advancement of environmental sustainability, economic prosperity, and social equity within the confines of Tiruchirappalli. By facilitating access to clean water, the proposed prototype is poised to catalyze improvements in public health outcomes, bolster livelihood opportunities, and foster the resilience of communities grappling with water-related challenges. Furthermore, the project underscores the pivotal significance of indigenous innovation and collaborative endeavors in confronting the intricate and far-reaching ramifications of global water scarcity.

As we embark upon this ambitious undertaking, our aspiration transcends the mere development of a functional prototype; rather, it encompasses the overarching ambition of catalyzing broader dialogues and catalytic actions geared towards the adoption of sustainable water management practices. By substantiating the feasibility and efficacy of solar-driven desalination technology within the context of Tiruchirappalli, our endeavor aspires to serve as a trailblazing exemplar, igniting a cascade of analogous initiatives in other water-stressed regions worldwide, and thereby effectuating a meaningful and enduring contribution towards the alleviation of water scarcity on a global scale.

MOTIVATION

The impetus behind this project is rooted in the urgent need to confront the pervasive challenges of water scarcity, particularly in urban settings like Tiruchirappalli. Despite its vibrancy, Tiruchirappalli grapples with significant obstacles related to water accessibility and quality, posing threats to public health and hindering social and economic progress.

At the core of our project lies a deep commitment to leveraging innovative solutions to address the complexities of water scarcity. Solar still technology emerges as a promising avenue, offering a sustainable and environmentally conscious approach to freshwater production by harnessing solar energy. Through the utilization of locally available resources and renewable energy sources, our project seeks to develop a solution that is not only accessible but also economically viable for Tiruchirappalli's communities.

Moreover, our project is driven by a steadfast dedication to promoting resilience and sustainability amidst the challenges of climate change and environmental degradation. By advocating for the widespread adoption of solar-driven desalination technology, we aim to reduce dependence on conventional water sources, which are increasingly vulnerable to climate-related disruptions.

Ultimately, our vision is to create a more water-secure future for the residents of Tiruchirappalli. By providing access to clean and potable water, we seek to enhance quality of life, bolster economic prosperity, and foster a society that is sustainable and equitable for all. Through collaborative efforts and innovative solutions, we strive to pave the way towards a future where every individual's basic needs are met with dignity and efficacy. Figures 1 and 2 illustrate the variations in solar intensity in the Trichy district, underscoring the significance of solar energy in our approach.

Methodology

The essence of our endeavor lies in the development of a water purification system harnessing the inexhaustible energy of solar radiation. Recognizing the abundant solar resources available, particularly in locales like Trichy, where sunlight is prevalent throughout the year, we aspire to strategically leverage this invaluable asset to address the pressing need for water purification.

Our vision encompasses the transformation of saline or impure water, sourced from various outlets such as sewage treatment plants, into potable water through judicious utilization of solar energy. Conventional methods like Reverse Osmosis (RO), while effective, entail substantial energy consumption, predominantly sourced from fossil fuels, exacerbating environmental concerns such as global warming and resource depletion. Thus, we are resolute in our commitment to develop a system that eschews reliance on fossil fuels, instead harnessing the clean and renewable energy of the sun. Central to our approach is the concept of autonomy, minimizing human intervention during system operation. By integrating advanced control mechanisms and sensor technologies, our system aims to operate seamlessly, optimizing water purification processes while requiring minimal oversight.

Moreover, our innovation extends beyond mere purification, as we aim to maximize resource utilization by repurposing brine water, a byproduct of the purification process. Through innovative techniques, we seek to mitigate wastage and promote sustainability by effectively utilizing this secondary resource.

Our objectives coalesce around four fundamental pillars:

1. **Efficiency Enhancement:** Through meticulous design and optimization, our system aims to augment purification efficiency and overall output, ensuring the effective utilization of solar energy to produce clean water.

2. **Autonomous Operation:** By integrating state-of-the-art automation technologies, our system endeavors to function autonomously, minimizing human intervention and ensuring reliable and continuous water supply.

3**. Waste Reduction:** By incorporating strategies for the utilization of brine water, our system strives to minimize waste generation, promoting sustainable water management practices.

4. **Cost-effectiveness**: Grounded in the principles of affordability and accessibility, our solution seeks to provide a cost-effective alternative to conventional purification methods, ensuring broad accessibility and scalability.

In essence, our project represents a convergence of innovative technology, sustainability principles, and societal impact, with the overarching goal of fostering a more sustainable and resilient water supply system for communities in Trichy and beyond.

Month	Lalgudi	Thuraiyur	Manapparai	Thottiyam
January	6.20	6.40	6.31	6.62
February	7.07	7.09	6.89	7.10
March	7.00	6.94	6.90	6.92
April	6.35	6.37	6.25	6.37
May	5.69	5.87	5.61	5.71
June	5.32	5.28	5.31	5.37
July	5.32	5.24	5.04	5.35
August	5.65	5.33	5.44	5.39
September	5.90	5.97	6.12	6.08
October	5.62	5.17	5.27	5.55
November	4.93	5.01	5.16	4.97
December	4.81	5.45	4.74	5.53
TOTAL	**5.82**	**5.84**	**5.75**	**5.91**

AVERAGE: 5.83

AVERAGE: 5.83

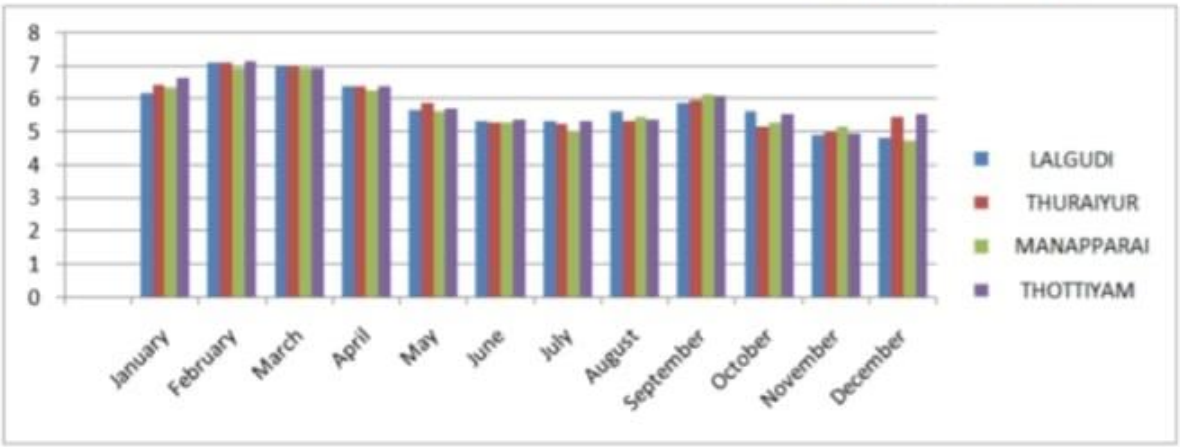


Figure 1 and 2: showing the solar intensity values of a year in Trichy district in kwh/m^2

DESIGN:

In our pursuit to design a solar still prototype that aligns with our specific requirements—ease of construction, maintenance, efficiency, and cost-effectiveness—we embarked on a meticulous exploration of existing research articles spanning various regions, including India and beyond.

Design Considerations:

Drawing upon insights gleaned from our comprehensive review of research literature, we embarked on the task of designing an innovative solar still solution. Our design considerations prioritized efficiency, ease of construction, and durability.

After scrutinizing various research findings, we arrived at a design that maximizes efficiency while remaining practical for construction and maintenance. Central to this design is a box structure created using SolidWorks software, meticulously crafted to optimize raw material utilization. The dimensions of this box are standardized at 80*50*45 cm^3, a configuration aimed at maximizing material efficiency without compromising structural integrity.

To enhance solar radiation absorption, we incorporated an inclined surface within the box structure, set at an optimal angle of 20 degrees relative to the horizontal plane. This inclination serves to maximize the surface area exposed to sunlight, thereby augmenting the amount of radiation trapped within the solar still.

Addressing concerns related to heat absorption by the plywood components, we devised a solution to mitigate this effect. By lining the interior surfaces of the box with reflective materials, we minimize heat absorption, thereby enhancing overall efficiency.

Furthermore, we prioritized durability by applying appropriate coatings to the exterior surfaces of the solar still. These coatings serve to mitigate potential damage from environmental factors such as moisture and termites, thereby prolonging the operational lifespan of the prototype.

The operational functionality of the solar still is enhanced through the integration of a piped system for water input, brine water removal, and collection of purified water. This piping infrastructure ensures efficient water flow management within the system, facilitating the purification process.

Prototype Construction:

For the construction of the solar still prototype, we meticulously selected a combination of materials to ensure durability, functionality, and cost-effectiveness. The chosen materials, along with their dimensions and applications, are as follows:

1. **4ft*6ft plywood of 12mm thickness**: This robust plywood serves as the primary structural component, forming the shell of the solar still. Through precise cutting and joining techniques, we crafted a sturdy box structure to contain the water purification process.

2. **19 in *17 in glass of 4mm thickness**: The glass panel, with its appropriate dimensions and thickness, functions as the transparent cover of the solar still. This component allows sunlight to penetrate the enclosure while preventing heat loss and facilitating the condensation process.

3. **Paint**: A weather-resistant paint coating was applied to the outer surfaces of the plywood shell to enhance durability and protect against moisture infiltration. This step ensures the longevity of the prototype under various environmental conditions.

4. **Aluminium foils**: Aluminium foils were utilized to line the inner surfaces of the plywood shell. This reflective material minimizes heat absorption, thereby optimizing the efficiency of the solar still by maximizing solar radiation absorption.

5. **Silicone pipes**: Silicone pipes were employed to facilitate the ingress and egress of water within the solar still. These flexible and durable pipes ensure efficient water flow management, enabling the collection of non-potable water and the removal of brine water.

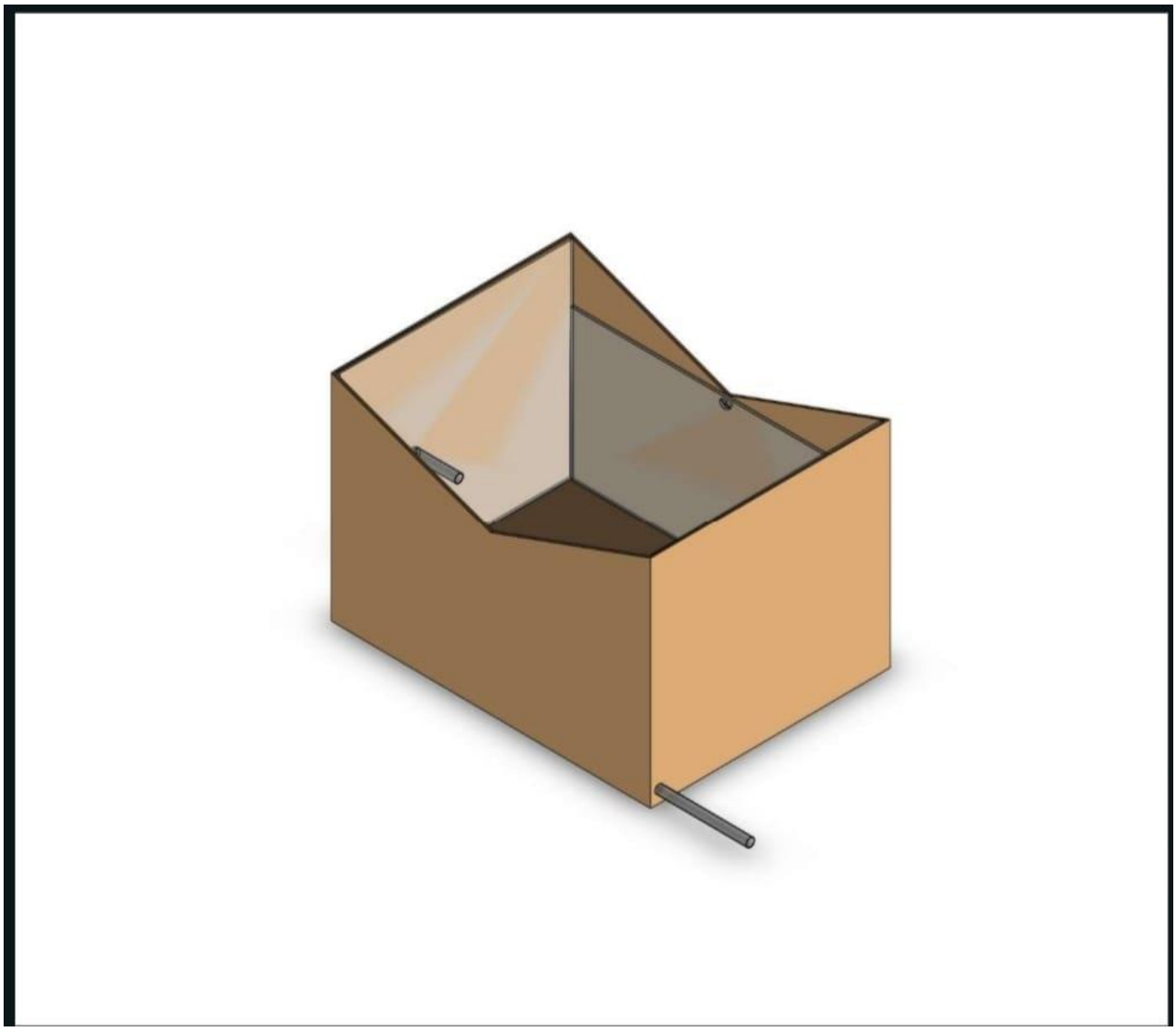
6. **PVC pipes**: Half-cut PVC pipes of 25mm diameter were strategically positioned within the shell to serve as conduits for transporting condensed water from the glass panel to the exterior. M-seal was utilized to seal these pipes securely, preventing any leakage.

Throughout the construction process, meticulous attention was paid to waterproofing measures to prevent any leakage. Provisions were also made within the shell to accommodate pumps, water level measuring sensors, and wiring connections, ensuring seamless integration of electronic components.

Furthermore, our commitment to sustainability led us to repurpose certain items from previous projects, minimizing waste and reducing manufacturing costs. This strategic reuse of materials not only aligns with our environmental ethos but also contributes to the overall affordability of the prototype.

In summary, the careful selection and utilization of materials, coupled with precise construction techniques, culminated in the creation of a solar still prototype that embodies durability, functionality, and cost-effectiveness, while also adhering to principles of sustainability.

Figure 3: 3D Design of the solar still



Electronics Control System Development

In order to facilitate autonomous operation and minimize human intervention in the functioning of the solar still, we devised a sophisticated electronic control system. This system is designed to monitor the water level within the still in real-time and initiate corrective actions as necessary, while also addressing the periodic need for brine water drainage. However, a key challenge we faced was ensuring a reliable power supply to sustain the system's operations continuously.

For the electronic components of our system, we carefully selected the following:

1. **Arduino Nano**: Serving as the central processing unit, the Arduino Nano collects real-time data from the water level measuring sensor and coordinates the operation of other components.

2. **2-channel 5V relay**: These relays act as switches, enabling the Arduino to control the activation of the submersible pumps based on the water level readings.

3. **9V submersible pumps**: These pumps are responsible for refilling the solar still when the water level falls below a specified threshold and for draining the brine water after a predetermined duration of operation.

4. **Water level measuring sensor**: This sensor provides real-time feedback to the Arduino regarding the water level within the solar still.

5. **ICR 18650 2200mAh 3.7V battery**: To ensure uninterrupted operation, we employed a custom-made lithium-ion battery pack consisting of two 3.7V batteries connected in series, providing a total voltage of 7.4V and a capacity of 2200mAh.

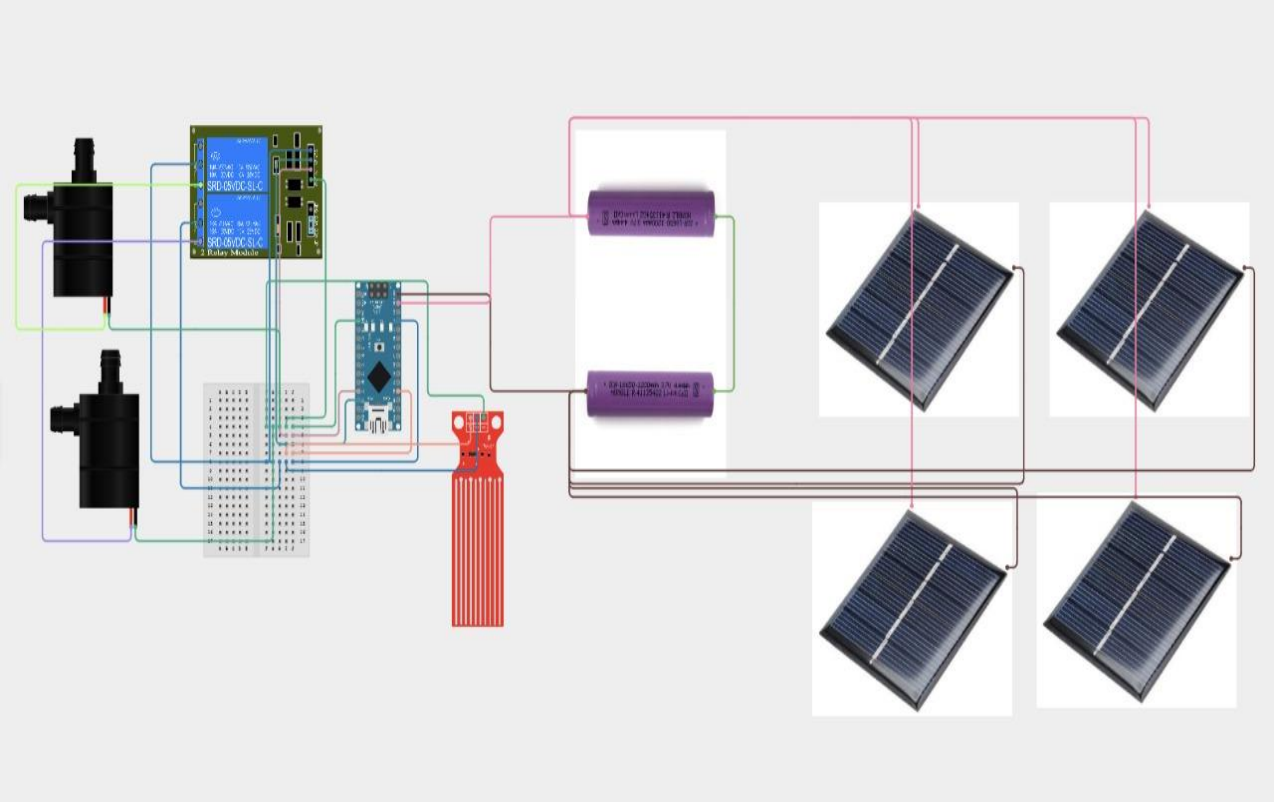
6. **6V 45mA Solar panel:** To maintain the battery charge, we utilized four 6V 45mA solar photovoltaic panels, ensuring a steady supply of power to the system.

The system operates as follows: When the water level falls below the specified threshold, the Arduino activates Pump 1 via the relay, replenishing the water until the desired level is reached. Additionally, after a predetermined operational duration (e.g., 12 hours), the Arduino triggers Pump 2 to drain the accumulated brine water from the solar still. This action is facilitated by the second channel of the relay, which controls the pump's operation.

To sustain the system's power requirements, the battery pack is continuously charged by the solar panels. The Arduino regulates the charging process to safeguard battery health, ensuring long-term reliability.

In summary, our electronic control system enables the autonomous operation of the solar still, optimizing water level management and brine water drainage while maintaining a sustainable power supply through solar energy harvesting and battery management protocols.

Figure 4: showing the circuit diagram of the control system



ARDIUNO CODING:

```
// Include the necessary libraries
#include <Arduino.h>
// Define pin numbers
const int water Level Sensor Pin = A0; // Analog pin for water level sensor
const int pump1RelayPin = 9;        // Digital pin for pump 1 relay control
const int pump2RelayPin = 10;       // Digital pin for pump 2 relay control
void setup () {
  // Set the relay pins as outputs
  pin Mode (pump1RelayPin, OUTPUT);
  pin Mode (pump2RelayPin, OUTPUT);
  // Set up serial communication
  Serial.begin (9600);
}
void loop() {
  // Read the analog value from the water level sensor
  int water Level = analog Read (water Level Sensor Pin);
 // Print the analog value for debugging
  Serial.print ("Water level: ");
  Serial.println (water Level);
 // Turn on pump 2 for 1800 seconds every 84600 seconds
  unsigned long current Time = millis () % 84600000; // Get current time in
milliseconds modulo 84600 seconds
  if (current Time >= 0 && current Time < 1800000) {
    digital Write (pump2RelayPin, HIGH); // Turn on pump 2
    digital Write (pump1RelayPin, LOW);  // Turn off pump 1 when pump 2 is
running
    Serial.println("Pump 2 is running. Pump 1 turned off.");
```

```
    delay(1800000); // Wait for 1800 seconds (30 minutes) before turning on
pump 1
  } else {
    digital Write (pump2RelayPin, LOW); // Turn off pump 2 after 1800 seconds
    // If it's not pump 2 operation time, run pump 1 normally based on water
condition
    if (water Level > 500) {
      digital Write(pump1RelayPin, LOW); // Turn off pump 1
      Serial.println("Water level is high. Pump 1 turned off.");
    } else {
      digitalWrite(pump1RelayPin, HIGH);
      Serial.println("Water level is low or normal. Pump 1 turned on.");
    }
  }

  // Wait for 84600 seconds before restarting the cycle
  if (current Time >= 84598500) {
    digital Write (pump2RelayPin, LOW); // Turn off pump 2 before restarting
the cycle
    delay (84600000 – current Time); // Wait for the remaining time in the cycle
  }

  // Delay for stability
  Delay (1000); // Check water level and pump operation every second
}
```

Cost Analysis:

Solar Still:

1. Plywood: Rs. 1300
2. Glass: Rs. 380
3. Paints, glue, etc.: Rs. 150
4. Aluminium foil: Rs. 70
5. Pipes: Rs. 150

Total Cost of Solar Still: Rs. 2050

Electronic Control System:

1. Arduino Nano: Rs. 250
2. 2-channel relay: Rs. 80
3. Pumps: Rs. 180
4. Mini breadboard: Rs. 30
5. Battery: Rs. 150
6. Sensor: Rs. 50
7. Wires: Rs. 50
8. Solar panel: Rs. 250

Total Cost of Electronic Control System: Rs. 1040

Overall Cost:

Total Cost of Solar Still + Total Cost of Electronic Control System:

= Rs. 2050 + Rs. 1040

= Rs. 3090

Therefore, the total estimated cost for the solar still and its electronic control system is Rs. 3090.

RESULTS:

Observations:

The phenomenon of fog formation beneath the glass surfaces occurs rapidly, typically within a span of around 300 seconds after initiating the solar still operation. This occurrence is a result of the condensation process, where moisture in the air condenses onto the cooler glass, forming visible droplets.

The initial collection of water droplets within the solar still system begins gradually and continues over an extended period, often taking up to two hours to accumulate the first measurable droplet. This timeframe reflects the process of moisture build up and condensation within the still.

The peak efficiency of water collection, amounting to approximately 70 ml, is achieved during the midday hours between 12:30 PM and 1:30 PM. This timeframe correlates with the optimal conditions for solar radiation and heat accumulation, facilitating increased evaporation and condensation rates.

In the context of water usage, the solar still requires a substantial volume of input water—around 20liters —to yield a relatively smaller but concentrated output of approximately 500 ml. This ratio underscores the concentration effect achieved through the distillation process within the solar still, where impurities are left behind as fresh water evaporates and condenses.

The incorporation of jute bags into the solar still design contributes significantly to enhancing the rate of evaporation. These bags serve as effective wicks that draw water from the reservoir, promoting continuous evaporation and condensation cycles.

Similarly, the use of clinker stones within the solar still system further enhances evaporation rates. Clinker stones, with their porous and heat-retentive properties, create an environment conducive to increased surface area for evaporation, thereby facilitating greater water vapor production.

The integration of an electronic control system within the solar still setup reduces the reliance on manual labour. This electronic control system automates key operational processes, optimizing efficiency and minimizing the need for constant human intervention. Consequently, the solar still operates with

enhanced reliability and efficiency, making it more accessible and practical for various applications, particularly in remote or resource-constrained areas.

Efficiency calculation:

Intensity:

Intensity, in physics, specifically refers to the amount of energy transmitted through a unit area in a specific direction per unit time. It is a fundamental concept used to describe the strength of various physical phenomena, including waves such as light, sound, and electromagnetic radiation.

The formula for intensity varies depending on the type of wave being considered:

1. **Sound Waves**:

In the case of sound waves, intensity (I) is defined as the power (P) per unit area (A) through which the sound is passing. The formula for sound intensity is:

$$I = P/A$$

where:

- I is the intensity of the sound wave (measured in watts per square meter, W/m^2).

- P is the power of the sound wave (measured in watts, W).

- A is the area through which the sound wave is passing (measured in square meters, m^2).

2. **Light Waves**:

In the context of electromagnetic waves like light, intensity refers to the radiant flux (power) per unit area perpendicular to the direction of propagation. The formula for light intensity can be derived from the energy E carried by the wave per unit time t through a surface area A:

$$I = E/At$$

Alternatively, intensity can also be expressed in terms of the electric field E and the magnetic field B of the electromagnetic wave:

$$I=1/2€_0 CE^2$$

where:

- I is the intensity of the light wave (measured in watts per square meter, W/m^2).

- E is the amplitude of the electric field (measured in volts per meter, V/m).

- ϵ_0 is the vacuum permittivity 8.85×10^{-12} F/m.

- C is the speed of light in vacuum 3×10^8 m/s.

The **zenith angle**, also known as the solar zenith angle or simply zenith angle, is an important concept in astronomy, atmospheric science, and solar energy studies. It is defined as the angle between the vertical (i.e., the direction directly above a location) and the line joining the observer to the sun. The zenith angle is measured from directly overhead (the zenith, at 90 degrees) to the horizon (0 degrees).

In more technical terms:

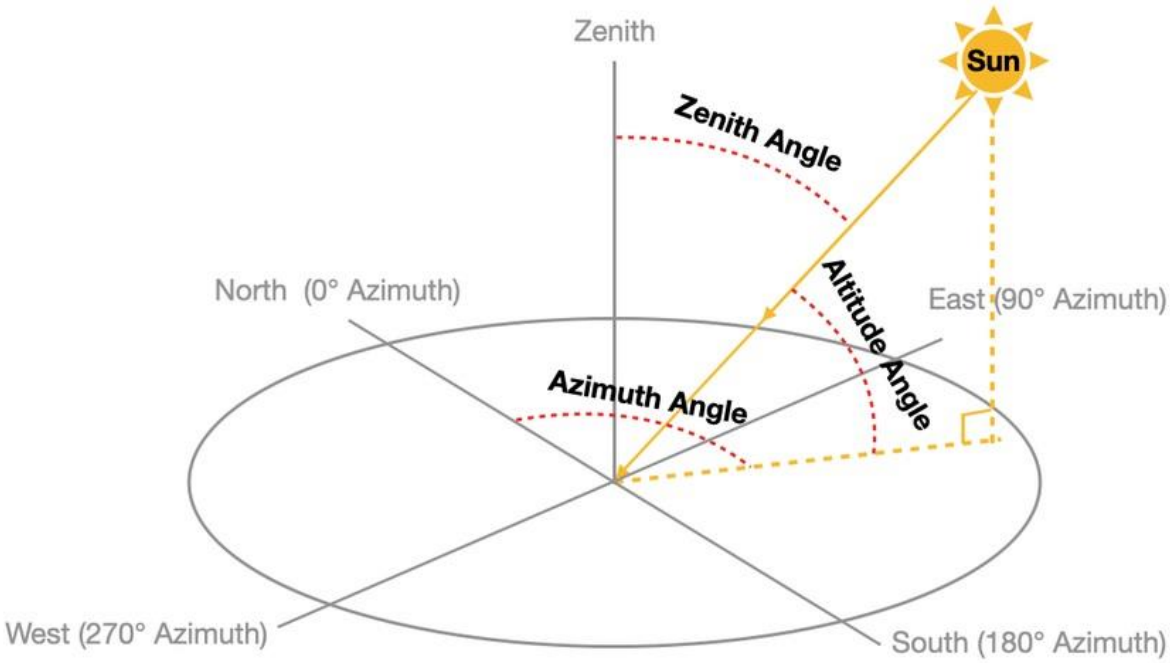
- The zenith angle θ is calculated based on the position of the sun relative to an observer on the Earth's surface.

- The zenith angle is measured from the vertical (zenith) downward to the point where a line from the observer to the sun intersects the celestial dome.

The zenith angle changes throughout the day as the position of the sun in the sky changes due to the Earth's rotation. At sunrise and sunset, the zenith angle is 90 degrees (horizon), while at solar noon (when the sun is highest in the sky), the zenith angle is 0 degrees (directly overhead).

The zenith angle affects the amount of atmosphere through which sunlight passes before reaching the Earth's surface. A high zenith angle (sun low in the sky) means that sunlight must pass through more atmosphere, which can affect the intensity and quality of light reaching the surface. This angle is important in solar energy applications because the amount of solar radiation received at the Earth's surface is influenced by the zenith angle; a smaller zenith angle (sun closer to zenith) implies more direct and intense solar radiation.

Figure 5: showing the zenith angle , altitude angle and azimuthal angle



$$I = k.\cos\theta$$

where;

k is the solar intensity of the earth which is constant of 1380 w/m^2 above the atmosphere at equator.

θ is the zenith angle at a particular place.

I is the solar intensity at a particular.

The hourly and over all efficiency of solar still(η) calculated using the equation.

$$\eta = \frac{\sum m \times L_v}{\sum I \times A}$$

where m is the distillate collected, I is the average solar radiation intensity, A is the basin area and L_v is the latent heat of vaporization of water.

Mass of the water collected = 0.5 kg

Latent heat of Vaporization= 2257 kJ

Average solar Intensity= 480 W/m^2

Area = 0.38 m^2

$$\eta = \frac{0.5\ kg \times 2257\ kJ}{480\ W/m^2 \times 0.38 m^2} = \mathbf{21.48\%}$$

Conclusions:

The completion of this project marks a significant milestone in our journey towards harnessing the resources of our surroundings for the betterment of society. Through our efforts, we have endeavored to provide a cost-effective solution for purifying water using the abundant solar energy available in the city of Trichy. We are optimistic that our project will make a meaningful contribution to addressing the pressing issue of water scarcity in our community.

Throughout the course of this project, we have not only achieved our intended goals but also gained valuable insights that will be instrumental in shaping our future endeavors. Our journey has been characterized by learning, innovation, and a deepened understanding of the intricate challenges associated with water purification and renewable energy utilization.

As we reflect on our achievements, we are filled with gratitude for the opportunity to contribute to the betterment of our society. We are committed to leveraging our newfound knowledge and experiences to continue developing impactful solutions that address the needs of our community and beyond.

Moving forward, we remain dedicated to the principles of sustainability, innovation, and social responsibility. With each project, we strive to push the boundaries of possibility and create a brighter, more resilient future for all.

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- Journal: Journal of Chemical and Pharmaceutical Sciences, ISSN: 0947-2115, Volume [Volume Number], Issue [Issue Number], Pages [Page Range]

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